STAFF REVIEW · H446 304 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Weighted contributions from links, damping and iterative calculation.

2. Required virtual memory page not resident in RAM.

3. Rows matching the join condition.

4. mid3 (7) then mid5 (11) found; index5.

I Do — source worked steps

1. Predict first — Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.

2. Check reasoning — Merge sort has O(n log n) worst-case growth and generally needs O(n) auxiliary memory.

We Do — guided CHECK

1. Predict first
Identify exactly which implementation state is wrong, e.g. high never decreases.
Check reasoning
Set a small test with a target below the midpoint to check the high-bound update independently.

You Do — source mark schemes

1. PC next instruction address [1]; MAR address [1]; MDR transferred value [1]; CIR current instruction [1]; ACC intermediate result [1]; address/data/control bus role [1].

2. mid3 value8 [1], low4 [1], mid5 value12 [1], return index5 [1], O(log n) [1] from halving [1].

3. Check data quality/incompleteness [2]; privacy and lawful use [2]; human oversight and appeal/false-positive impact [2].

Exit Ticket — source answer

1. False. An unseen trace or solution provides better evidence.